IAN PORTO

(647) 575 - 4364 ◇ Toronto, ON
ianporto2003@gmail.com ◇ linkedin.com/in/Ian-Porto ◇ IanPorto.ca

EDUCATION

M.Sc. Student - Earth and Space Sciences, Lassonde School of Engineering — Sep 2025 - Present
Queen Elizabeth II Graduate Scholarship in Science and Technology — Sep 2026
Enbridge Graduate Student Award — Sep 2026
Mitacs Globalink Research Award Abroad — Jan 2026
York Graduate Scholarship — Sep 2025

Bachelor of Engineering - Space Engineering, Lassonde School of Engineering — Sep 2021 - Jun 2025
Graduated First Class With Distinction. Member of Dean's Honour Roll.
Lassonde Undergraduate Research Award ×2 — Apr 2023 & 2024
Lassonde Undergraduate Research Award Fall/Winter ×2 — Sep 2022 - Apr 2025

EXPERIENCE

Research Assistant — Feb 2023 - Present
Nanosatellite Research Lab. — *Toronto, ON*
Under the supervision of Dr. Regina Lee

- With a research focus on Very Short Arc Initial Orbit Determination
- Work in Space Engineering, performing various simulations, tests, characterizations, and design.

Teaching Assistant - Multiple Courses — Sep 2025 - Present
Lassonde School of Engineering & York University — *Toronto, ON*

- Engineering Capstone Project (ENG 4000) (Sep 2026 - Apr 2027)
- Space Hardware (ESSE 4350) (Sep 2026 - Apr 2027)
- Renaissance Engineering 2 (ENG 1102) (Jan 2027 - Apr 2027)
- Renaissance Engineering 1 (ENG 1101) (Sep 2025 - Dec 2025)
- Computational Methods for Space Engineers (ESSE 2220) (Sep 2025 - Dec 2025)

Research Assistant & International Liason — Jan 2026 - Aug 2026
Universidad Politécnica de Madrid, Instituto Universitario de Microgravedad "Ignacio Da Riva" — *Madrid, ES*
Under the supervision of Dr. Regina Lee, Dr. Angel Porras-Hermoso and Dr. Elena Roibas Millan

- Assist with the development of the UPMSAT-4 mission
- Manage relations between international academic institutions ensuring that all stakeholders are satisfied with mission progress.
- Work as the Project Manager for York University's payload onboard the satellite.

Space Engineering Ambassador (Recruitment & Campus Ambassador Lead) — Apr 2024 - Jun 2025
Lassonde School of Engineering — *Toronto, ON*

- Facilitated student transitions by conducting campus tours, acting as a peer role model, and providing guidance on navigating university procedures and resources.
- Managed digital and physical outreach by developing website content, social media strategies, and promotional materials to increase event awareness and student engagement.

PUBLICATIONS

- **I Porto**, M Myhre, S Mesón-Pérez, A Porras-Hermoso, G Sohn, R SK Lee, Hardware-in-the-Loop Validation of Physics-Informed Neural Network for Angles-Only Initial Orbit Determination in Very Short Arcs, 46th COSPAR Scientific Assembly 2026 (Accepted)
- M Myhre, **I Porto**, V Suthakar, R SK Lee, Shallow Neural Network Centroid Algorithm Comparison for a Dual-Use Star Tracker Payload for Space Situational Awareness, 46th COSPAR Scientific Assembly 2026 (Accepted)
- **I Porto**, M Myhre, V Suthakar, RSK Lee, Thermo-Optical Characterization and Qualification of a Space Situational Awareness (SSA) Payload, 5th IAA Conference on Space Situational Awareness.

- M Myhre, **I Porto**, V Suthakar, R SK Lee, Centroid Algorithm Comparison in a Dual-Use Star Tracker for Space Situational Awareness.
- V Suthakar, **I Porto**, SKM Dasanayaka, RSK Lee, MI Robert, R Olteanu, A Bucur, M Kübler, JI Hubbard, M Scherillo, F Newland, V Braun, Stratospheric Observation of Resident Space Objects using Photometric and Infrared Sensors. 9th European Conference on Space Debris,
- V. Suthakar, **I. Porto**, M. Myhre, A. A. Sanvido, R Clark, & R. S. K. Lee, RSONAR: Data-Driven Evaluation of Dual-Use Star Tracker for Stratospheric Space Situational Awareness (SSA), MDPI Sensors.
- **I. Porto**, T. Barrett, L. Gierull R. Qashoa, R. Clark, R. S. K. Lee, (August 2023). Solving the Degeneracy Issue in Finding Keplerian Elements Using Unscented Kalman Filters. Lassonde Undergraduate Research Conference.

CERTIFICATES

- ESA Academy - Space Standards Training Course — Jun 2026
- STK Grand Master — Apr 2026
- STK Master Integration — Apr 2026
- Project Management Foundations — Sep 2024
- Advanced Certificate of Proficiency in Amateur Radio, Callsign VA3PZZ — Dec 2023

PROJECTS

RSONAR-5b: I am currently a main payload lead, and systems engineering lead for RSONAR-5b, which will fly from Palmas, Brazil, in Q3 2027

UPMSAT-4, YU-MUST Payload: I am currently working as the project manager on the York University Multi-Use Star Tracker payload which will fly onboard Universidad Politécnica de Madrid's satellite, UPMSAT-4.

Space Engineering Alumni Event: I presented alumni from various fields including the Canadian Space Agency, MDA Space, and more. I focused on keeping a large audience engaged, and developed the presentation to ensure active participation from all of the Alumni.

REDWING Microsatellite: A Canadian R&D microsatellite for Space Domain Awareness. In this project, I work on verifying and validating orbits and attitudes for the REDWING Microsatellite while working closely alongside our industry partner (Magellan Aerospace)

DAVINCI-360 (Capstone Project): A sub-Payload of the York University's Nanosatellite Lab's R3 Mission, launched on the CSA's STRATOS 2025 Mission. The mission took 360° images of the sunset and night sky.

ESSENCE CubeSat: A project funded by the CSA's Canadian CubeSat Project. Worked under the Systems sub-team to facilitate everything from critical design review to Integration and Systems testing. Launched in June 2023 on SpaceX CRS-28

OUTREACH & VOLUNTEERING

- **Space Engineering Gateway Events**, Various Events — Sep 2024 - Present
- **Summer Camp Presentations**, Aurora, Ontario — Aug 2024
- **This Is Lassonde Podcast** — Jan 2024
- **Lassonde Undergraduate Research Award Video Series** — Jul 2023 - Aug 2023
- **Orientation Captain**, Lassonde Week0, Lassonde School of Engineering — Jun 2023 - Sep 2023
- **Highschool Outreach Presentation** — Jun 2023
- **Satellites and You**, Ontario Science Center — Mar 2023
- **Store Manager**, Lassonde Engineering Society — Sep 2022 - Apr 2023
- **Orientation Leader**, Lassonde Week0, Lassonde School of Engineering — Aug 2022